

Both transmission channels modulated by $(1 - \epsilon \cdot \bar{P})$

SD layer (information, behaviour)

saliency: $S(t) = \min(1, (I/N) \cdot \alpha \cdot e^{(-\lambda t/100)})$

risk: $R_i(t) = (1 - \eta) \cdot R_{\text{target}} + \eta \cdot R_i(t-1)$

protection: $P_i = \sigma(\kappa(R_i - \mu))$, $\bar{P} = \text{mean } P_i$

saliency $S(t)$
risk targets ↓

aggregate (S, E, I, R)
mean protection $\bar{P}$ ↑

ABM layer (individuals)

agents with states $S / E / I / R$

stochastic transitions; $\bar{P}$ in both channels

infected $I(t)$ ↓

risk_m ↑ admissions ↓

beds / ICU ↑

Mosquito layer

humans $\rightarrow E_m \rightarrow I_m \rightarrow$ humans

incubation 10 d, lifespan 14 d

risk_m = $b \cdot p_{mh} \cdot (I_m/M)$

DES layer (hospital)

beds / ICU / testing queues

deferred admissions (counted once)

capacity constraints

modulation

human:

$\beta(1 - \epsilon \cdot \bar{P}) \cdot I/N$

mosquito:

$\text{risk_m} \cdot (1 - \epsilon \cdot \bar{P})$